

CYNTERA

Drug Repurposing AI Report

Drug: Dapagliflozin
Disease: Heart Failure
Generated: 2026-08-13 16:51 UTC

Recommendation: NOT_RECOMMENDED

Executive Summary

CYNTERA v2.0 analysis of Dapagliflozin → Heart Failure produced a recommendation of 'NOT_RECOMMENDED'. Evidence Strength: 8.1% (LOW) | Mechanistic Plausibility: 49.0% (LOW) | Risk Level: 93.3% (HIGH). 1 claim(s) extracted from literature (1 from pubmed), 0 contradiction(s) detected. Prior knowledge: regulatory NONE (0%), repurposing EMERGING, mechanistic MODERATE, clinical HUMAN_EVIDENCE, knowledge maturity SPECULATIVE. Safety grade: D ■ Boxed warning detected. 2 mechanistic path(s) traced.

Three-Dimensional Score Assessment

Dimension	Score	Level	Details
Support Score (SS)	0.081	LOW	1 evidence records
Mechanistic Score (MS)	0.490	LOW	1 pathways
Risk Score (RS)	0.933	HIGH	0 failed trials, 0 contradictions

Scientific Context — Prior Knowledge

Dimension	Status	Confidence	Evidence
Regulatory	NONE	0%	No ChEMBL indication match found for this disease
Repurposing	EMERGING	50%	Pair-scoped literature exists for this disease
Mechanistic	MODERATE	49%	Validated path: Dapagliflozin → SLC5A2 (P31639) [inhibitor (ChEMBL)] →
Clinical	HUMAN_EVIDENCE	70%	Human clinical literature (RCT / meta-analysis / observational) for this pair
Knowledge Maturity	SPECULATIVE	0%	No prior knowledge cache entries for this pair

Candidate Biological Mechanisms

Candidate Mechanism 1: Mechanism 1: SLC5A2 (P31639) [WEAK_SPECULATIVE]

Drug: Dapagliflozin → Target: SLC5A2 (P31639) → Gene: SLC5A2 → Disease: Heart Failure

- Drug: Dapagliflozin —**INHIBITOR**→ Target: SLC5A2 (P31639) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: SUPPORTED; Source: ChEMBL

- Target: SLC5A2 (P31639) —**ENCODED_BY_DISEASE_ASSOCIATED_GENE**→ Gene: SLC5A2 [\[Open UniProt\]](#) [\[Open Targets Target\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

- Gene: SLC5A2 —**ASSOCIATED_WITH**→ Disease: Heart Failure [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Candidate Mechanism 2: Mechanism 2: SLC5A2 (P31639) via Transport of glucose and other sugars, bile salts and organic acids, metal ions and amine compounds (R-HSA-163200) [WEAK_SPECULATIVE]

Drug: Dapagliflozin → Target: SLC5A2 (P31639) → Pathway: Transport of glucose and other sugars, bile salts and organic acids, metal ions and amine compounds (R-HSA-163200) → Gene: SLC5A2 → Disease: Heart Failure

- Drug: Dapagliflozin —**INHIBITOR**→ Target: SLC5A2 (P31639) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: SUPPORTED; Source: ChEMBL

- Target: SLC5A2 (P31639) —**PARTICIPATES_IN**→ Pathway: Transport of glucose and other sugars, bile salts and organic acids, metal ions and amine compounds (R-HSA-163200) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Pathway: Transport of glucose and other sugars, bile salts and organic acids, metal ions and amine compounds (R-HSA-163200) —**CONTAINS_ASSOCIATED_GENE**→ Gene: SLC5A2 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

- Gene: SLC5A2 —**ASSOCIATED_WITH**→ Disease: Heart Failure [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: SLC5A2 (P31639) → Pathway: Transport of glucose and other sugars, bile salts and organic acids, metal ions and amine compounds (R-HSA-163200) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Transport of glucose and other sugars, bile salts and organic acids, metal ions and amine compounds (R-HSA-163200) → Gene: SLC5A2 is not supported by a mapped claim.

Biomedical Data Sources Accessed

Source Name	Status	Direct Source Portal
ChEMBL	SUCCESS	[Open ChEMBL]
UniProt	SUCCESS	[Open UniProt]
Reactome	SUCCESS	[Open Reactome]
Open Targets	SUCCESS	[Open Open Targets]
DisGeNET	SUCCESS	[Open DisGeNET]
PubMed	SUCCESS	[Open PubMed]
Europe PMC	SUCCESS	[Open Europe PMC]
ClinicalTrials.gov	SUCCESS	[Open ClinicalTrials.gov]

Recommendation Rule Engine Output

- Rule 0 (SAFETY VETO): ■ Boxed warning / contraindication detected. Risk Score = 0.933. Safety grade: D. NOT RECOMMENDED due to unacceptable safety profile / disease contraindication.
- Evidence signals:
- [FAIL] Literature support: SS = 0.081 (LOW) from 1 records
- [PASS] Mechanistic plausibility: MS = 0.490 (LOW), 1 pathway(s)
- [FAIL] Safety/Risk acceptable: RS = 0.933 (HIGH)
- [PASS] Evidence consistency: No contradictions
- [PASS] Human clinical data: Available

Data Gaps Identified

- Sparse evidence base (1 records). Consider COMPREHENSIVE retrieval policy (−0.10 confidence penalty).
- No prior knowledge entries found for this drug-disease pair. This appears to be a genuinely novel repurposing hypothesis (−0.05 confidence penalty).

Evidence Synthesis & Dimensional Assessment

1. Mechanistic Assessment

Mechanistic Assessment: LOW (Score = 0.490).

Literature Grounding: DATABASE_ONLY.

Candidate Biological Mechanisms Discovered: 2.

Primary Mechanistic Chain: Drug: Dapagliflozin → Target: SLC5A2 (SLC5A2 (Sodium/glucose cotransporter 2) [UniProt: P31639]) [INHIBITOR via ChEMBL] → Gene: SLC5A2 [ENCODED_BY_DISEASE_ASSOCIATED_GENE via Open Targets / DisGeNET] → Disease: Heart Failure [ASSOCIATED_WITH via Open Targets / DisGeNET].

Rationale: Mechanistic Score (MS) derived from candidate mechanism quality. Best candidate support level: WEAK_SPECULATIVE. Discovered 2 candidate biological mechanism(s). Literature grounding: DATABASE_ONLY (1 claim(s)). Final MS = 0.490 (LOW).

2. Clinical Evidence Assessment

Clinical Evidence Assessment: LOW Literature Support (Score = 0.081).

Clinical Trial Status: NOT_ATTEMPTED.

Clinical Trial Findings: 0 trial(s) evaluated, 0 clinically significant failure(s).

Literature Support Rationale: Support Score (SS) from 1 evidence records.

Breakdown: Reviews/Observational: 1 records (avg ERW: 0.70)

Formula: $SS = 1 - \exp(-0.12 \times \text{quality_weighted_sum})$. Quality-weighted sum = 0.70 (raw ERW sum would be 0.70, capped to prevent inflation from low-quality records).

Final SS = 0.081 (LOW).

3. Safety & Risk Assessment

Safety & Risk Assessment: Grade D — Significant safety concerns — strong contraindications identified (Risk Score = 0.933, HIGH).

Boxed Warning: YES.

Risk Factors & Signal Analysis: Risk Score (RS) computed from clinically significant safety signals.

Trial analysis: 0 clinically significant failure(s); 0 administrative/external termination(s) excluded.

Risk factors: FDA Boxed Warning detected; Safety grade: D (grade penalty: 3.0).

Formula: $RS = 1 - \exp(-0.30 \times \text{raw_risk_burden})$. Raw burden = 9.00. Final RS = 0.933 (HIGH).

4. Final Synthesis

Synthesis & Recommendation: NOT_RECOMMENDED.

Regulatory Status: NONE (Term: N/A).

1. Mechanistic Plausibility: LOW (0.490)

2. Literature/Clinical Support: LOW (0.081)

3. Risk & Safety Profile: HIGH (0.933, Grade D)

Conclusion Summary: Rule 0 (SAFETY VETO): ■ Boxed warning / contraindication detected. Risk Score = 0.933. Safety grade: D. NOT RECOMMENDED due to unacceptable safety profile / disease contraindication.

Confidence Narrative

Overall confidence is estimated at 85% after accounting for data gaps (total penalty: -15%). Evidence quality: The analysis is based on 1 retrieved records representing limited or low-quality evidence. ERW values are capped per evidence tier to prevent inflation from low-quality records (e.g., reviews and hypotheses). Safety assessment: Grade D — Significant safety concerns — strong contraindications identified. A boxed warning was detected. Safety grade scale: A = Excellent | B = Acceptable | C = Caution | D = Unsafe. Prior knowledge: No ChEMBL indication data available and no cache entries found for Dapagliflozin → Heart Failure. Treating as a novel repurposing hypothesis. Evidence evaluated on its own

merits.